

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
 May - June 2023

MA1004-1 – DISCRETE MATHEMATICS & NUMERICAL METHODS

Max. Marks: 100

Duration: 3 Hours

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Matrix of a symmetric relation is:**
 - Invertible matrix
 - Skew symmetric matrix
 - Matrix whose all-diagonal entries are 1.
 - Symmetric matrix
- If $U = \{x | x \text{ is an even integer} \wedge 0 \leq x \leq 12\}$, then the bit string of the set $A = \{2, 6, 8, 10\}$ is:**
 - 0101110
 - 101110
 - 0001110001110
 - 00100010101000
- Which of the following logical argument is a tautology?**
 - $[(p \vee q) \wedge q] \rightarrow q$
 - $[(p \vee q) \wedge q] \rightarrow p$
 - $[(p \vee q)] \rightarrow p$
 - $p \rightarrow p$
- The permutation $p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 2 & 5 \end{pmatrix}$ on $A = \{1, 2, 3, 4, 5\}$ is:**
 - Cyclic permutation
 - Even permutation
 - Neither odd nor even permutation
 - Odd permutation
- A simple graph can have _____**
 - multiple edges
 - self loops
 - parallel edges
 - no multiple edges, self loops and parallel edges
- In adjacency matrix column represent**
 - Vertices
 - edges
 - paths
 - circuits
- The complete graph with four vertices has k edges where k is**
 - 3
 - 4
 - 5
 - 6
- A graph is bipartite if and only if its chromatic number is _____**
 - 1
 - 2
 - odd
 - even
- The first approximation to a real root of the equation $x^3 - 2x^2 - 2 = 0$ in $(2, 3)$ by Regula-Falsi method is**
 - 2.95
 - 2.22
 - 2.56
 - 2.09
- The method which approximates the curve to the tangent to find root of an algebraic equation is:**
 - Newton-Raphson method
 - Taylor's series method
 - Regula-Falsi method
 - Euler's method
- The Taylor's series method to find the solution of the given first order ordinary differential equation at $y(x_1)$ for given h is**
 - $y_1 = y_0 - \frac{h}{1!}y_0' + \frac{h^2}{2!}y_0'' - \frac{h^3}{3!}y_0''' + \dots$
 - $y_1 = \frac{h}{1!}y_0' + \frac{h^2}{2!}y_0'' + \frac{h^3}{3!}y_0''' + \dots$
 - $y_1 = y_0 + \frac{h}{1!}y_0' + \frac{h^2}{2!}y_0'' + \frac{h^3}{3!}y_0''' + \dots$
 - $y_1 = \frac{h}{1!}y_0' - \frac{h^2}{2!}y_0'' + \frac{h^3}{3!}y_0''' + \dots$
- Given the equation $x^3 + x - 3 = 0$, which of the intervals below contains a root?**
 - $[-1, 1]$
 - $[0, 1]$
 - $[2, 3]$
 - $[1, 2]$

13. The standard five point formula for solving Laplace's equation $u_{xx} + u_{yy} = 0$ is _____
- A) $u_{i,j} = \frac{1}{4}[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$ B) $u_{i,j} = u_0 + \frac{1}{4}[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$
 C) $u_{i,j} = \frac{1}{3}[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$ D) $u_{i,j} = \frac{1}{4}[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}] - u_{0,0}$
14. The difference equation to solve wave equation $u_{tt} = c^2 u_{xx}$ when $\alpha c = 1$ is _____
- A) $u_{i,j+1} = u_{i-1,j} + u_{i+1,j} + u_{i,j-1}$ B) $u_{i,j+1} = u_{i-1,j} + u_{i+1,j}$
 C) $u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1}$ D) $u_{i,j+1} = \frac{1}{2}(u_{i-1,j} + u_{i+1,j})$
15. The forward difference approximation for $\frac{\partial u}{\partial x}$ is _____
- A) $\frac{u_{i,j} - u_{i-1,j}}{h} + O(h)$ B) $\frac{u_{i+1,j} - u_{i,j}}{h} + O(h)$
 C) $\frac{u_{i+1,j} - u_{i-1,j}}{2h} + O(h^2)$ D) $\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + O(h^2)$
16. For the heat equation $u_t = \frac{1}{2} u_{xx}$, $h = k = 1$, the value of mesh ratio parameter α is _____
- A) 1 B) 2
 C) 0.25 D) $\frac{1}{2}$
17. The value of a_0 in the Fourier series of $f(x) = x$ in the interval $(0, 2\pi)$ is :
- A) $2\pi^2$ B) π
 C) $-\pi$ D) 0
18. If $f(x)$ is defined in $(-\pi, \pi)$, then a_n
- A) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$ B) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$
 C) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$ D) $\int_{-\pi}^{\pi} f(x) \cos nx dx$
19. If $F(x(t)) = X(\omega)$, is the Fourier transform of the function $x(t)$, then $F(x(t) \cos at) =$
- A) $\frac{1}{a} \{X(\omega + a) + X(\omega - a)\}$ B) $\frac{1}{2} \{X(\omega + a) - X(\omega - a)\}$
 C) $\frac{1}{a} \{X(\omega + a) - X(\omega - a)\}$ D) $\frac{1}{2} \{X(\omega + a) + X(\omega - a)\}$
20. The inverse DFT $x(n)$, of the sequence $X(k)$ is:
- A) $\frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{i \frac{2\pi}{N} kn}; 0 \leq n \leq N-1$ B) $\frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{i \frac{2\pi}{N} kn}; 0 \leq n \leq N-1$
 C) $\frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{i \frac{2\pi}{N} kn}; 0 \leq n \leq N-1$ D) $\frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{i \frac{2\pi}{N} kn}; 0 \leq n \leq N-1$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) i) Define Eulerian circuit and Hamiltonian circuit. Give an example of A) A graph which is Hamiltonian but not Eulerian. B) A graph which is Eulerian but not Hamiltonian. ii) Show that a cycle C_6 is bipartite. b) Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1,2), (2,3), (3,4), (2,1)\}$. Find the transitive closure of R using Warshall's algorithm. c) i) Determine whether the following permutation on $A = \{1,2,3,4,5,6,7\}$ is even or odd. $p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 4 & 5 & 7 & 6 & 3 & 1 \end{pmatrix}$. ii) Suppose that the universal set is $U = \{1,2,3,4,5,6,7,8,9,10\}$, $A = \{1,2,3,4,5\}$, $B = \{2,3,6,7\}$. Express $A \oplus B$ in bitstrings using bitstring rules only.	6 6 4	L1 L3 L3	2 1 1	1 2 2
2.	a) i) Define planar graph. Show that (A) K_5 is non planar (B) $K_{3,3}$ is non planar. ii) Define coloring and chromatic number of graphs. b) In terms of propositional variables p, q, r , state the rule of inference for: i) Modus ponens, ii) Modus tollens, iii) Law of the syllogism. c) i) Encode the message "THE TOMATO IS A PLANT IN THE NIGHT SHADE FAMILY" using the key word "TOMATO" by using keyword columnar transposition method. ii) Define floor function.	6 6 4	L1 L1 L3	2 1 1	1 1 2

3. a) Write adjacency and incidence matrix of i) $K_{2,3}$ ii) W_5 . 6 L3 2 2
 b) Let $A = \{1, 2, 3, 4\}$ and let R be the relation on A defined by xRy if and only if "x divides y", written $x | y$.
 i) Write down R as set of ordered pairs.
 ii) Draw the digraph of R
 iii) Determine the in-degree and out-degrees of the vertices in the digraph.
 iv) Determine whether R is reflexive, symmetric or transitive. 6 L3 1 2
 c) State handshaking theorem. Prove that undirected graph has an even number of vertices of odd degree. 4 L1 2 1

Unit – II

4. a) Evaluate the pivotal values of the equation $u_{tt} = 16u_{xx}$ taking $h=1$ up to $t=1.25$, the boundary conditions are $u(0, t) = u(5, t) = 0$; $u(x, 0) = x^2(5 - x)$ & $u_t(x, 0) = 0$. 6 L3 4 2
 b) Find the solution of the parabolic equation $u_{xx} = 2u_t$ when $u(0, t) = u(4, t) = 0$, & $u(x, 0) = x(4 - x)$ taking $h=1$. Find the values up to $t = 5$. 6 L3 4 2
 c) Use the method of false position to find the root of the equation $x \log_{10} x - 1.2 = 0$ in $(2, 3)$. Carry out three iterations. 4 L3 3 2
 5. a) Using Modified Euler's method find an approximate value of y , when $x = 0.1$, taking $h = 0.05$ for $y' = x^2 + y$; $y(0) = 1$. 6 L3 3 2
 b) Solve $\nabla^2 u = -81xy$, $0 < x, y < 1$, $h = 1/3 = k$, $u(0, y) = 0 = u(x, 0)$, $u(1, y) = 100 = u(x, 1)$. 6 L3 4 2
 c) Show that Newton-Raphson method has second order convergence. 4 L1 3 1
 6. a) Solve the Laplace equation $u_{xx} + u_{yy} = 0$ for $0 < x < 1$, $0 < y < 1$. Given that $u(x, 0) = u(0, y) = 0$, $u(x, 1) = 6x$; $0 < x \leq 1$ and $u(1, y) = 3y$, $0 < y < 1$. Divide the region into 9 square meshes. 6 L3 4 2
 b) Using fourth order Runge-Kutta method solve the following initial value problem: $y' = x + y^2$, $y(0) = 1$. Find $y(0.2)$ taking $h=0.1$. 6 L3 3 2
 c) Using Taylor's series method compute y correct to 3 decimal places, at $x = 0.1$, given $y' = x - y^2$, $y(0) = 1$, $h=0.1$. 4 L3 3 2

Unit – III

7. a) Obtain the Fourier series of $f(x) = e^{-x}$, $0 < x < 2\pi$. $f(x + 2\pi) = f(x)$. 6 L3 5 2
 b) Find the Fourier cosine transform of $f(x) = \begin{cases} x, & \text{for } 0 < x < 1 \\ 2 - x, & \text{for } 1 < x < 2, \\ 0, & \text{for } x > 2 \end{cases}$ 6 L3 5 2
 c) Compute the N-point DFT of the signal given by, $x(n) = a^n$; $0 \leq n \leq N - 1$ 4 L3 5 2
 8. a) Find the Fourier series of $f(x) = \begin{cases} 1 + \frac{2x}{\pi} & (-\pi, 0) \\ 1 - \frac{2x}{\pi} & (0, \pi) \end{cases}$, $f(x + 2\pi) = f(x)$ 6 L3 5 2
 b) Find the half range Fourier cosine series of $f(x) = x^2$ in $0 \leq x \leq \pi$. 6 L3 5 2
 c) Show that i) $F_s[x f(x)] = -\frac{d}{ds} \{F_c(s)\}$
 ii) $F_c[x f(x)] = \frac{d}{ds} \{F_s(s)\}$. 4 L1 5 1